

# Hazem Hassan

Senior AI/ML Engineer — Computer Vision · LLM & Agentic Systems · Production ML

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## PROFESSIONAL SUMMARY

Senior ML Engineer / AI Technical Lead with 9+ years shipping production AI systems end-to-end across two tracks: computer vision & video analytics (OCR, multi-object tracking, action recognition, edge deployment on NVIDIA Jetson) and LLM-powered systems (agentic workflows with LangGraph, RAG with LlamaIndex, and a full evaluation/observability stack with Ragas and LangSmith). Strong software-engineering foundation — production Python, FastAPI/REST design, Docker, Kubernetes, CI/CD, ONNX/TensorRT — with a habit of thinking in latency, cost-per-query, and output-quality metrics. Comfortable owning the full lifecycle: architecture, code, deployment, evaluation, and monitoring.

## ROLES AND SPECIALTIES

**Roles:** Senior ML Engineer, AI Technical Lead, Senior Computer Vision Engineer, LLM/Agentic Systems Engineer.  
**Specialties:** Computer Vision Pipelines, OCR, Video Analytics, Multi-Object Tracking, Action Recognition, Agentic Workflows, RAG, LLM Evaluation, Automated Labeling, Human-in-the-Loop Workflows, Active Learning, Weak Supervision, Dataset Quality Assurance, ML APIs, MLOps, Cloud/Edge Deployment.

## CORE SKILLS

**Programming:** Python (expert), C/C++ (intermediate), CUDA / NVIDIA platforms, Git, Linux.  
**LLM & Agentic Systems:** LangGraph (agent workflows/ tool calling), LlamaIndex (RAG, embedding fine-tuning, retrieval orchestration), Multi-RAG, Embeddings, Reranking, Chunking Strategies, FAISS, Conversation Orchestration, NeMo Guardrails, Human-in-the-Loop/Escalation Routing, Prompt Engineering.  
**LLM Evaluation & Observability:** Ragas (retriever/generator eval), LangSmith (tracing + continuous evaluation), MT-Bench (LLM-as-judge), AlpacaEval, LM-Eval Harness, Elo Rating, Perplexity, Concurrency/Latency Benchmarking.  
**Computer Vision & NLP:** Object Detection, Segmentation, OCR, Video Analytics, Multi-Object Tracking, Action Recognition, Anomaly Detection, Face Recognition, Liveness Detection, Image Processing, Optical Flow, Diffusion Models (Image Generation/Editing, Controllable Synthesis).  
**ML/DL:** PyTorch, TensorFlow, Scikit-learn, Model Training & Fine-Tuning, Hyperparameter Tuning, Custom Loss Design, Evaluation Metrics, Error Analysis.  
**Data Engineering:** SQL, PostgreSQL, NoSQL Basics, Pandas, NumPy, Data Cleaning/Transformation, Large-Scale Image/Video/Text Processing, Dataset Preparation Pipelines, Feature Engineering.  
**Deployment & MLOps:** Model Serving, FastAPI, REST APIs, Docker, Kubernetes, CI/CD, Model Optimization, ONNX, TensorRT, Inference Performance Tuning, Monitoring & Failure Analysis.  
**Cloud & Edge:** GCP (Vertex AI), AWS (SageMaker / Ground Truth), NVIDIA Jetson, Real-time Inference, Low-latency Video Pipelines.  
**Leadership:** Technical Leadership, System Design, Cross-functional Collaboration, Mentoring, Code Reviews.

## WORK EXPERIENCE

### Senior ML Engineer

Beyond AI March 2024 — Present, Remote, EMEA  
--LLMs

- Built agentic LLM workflows with LangGraph — modeling agents as directed graphs with message passing, tool-calling, and execution orchestration across nodes — plus custom orchestration and RAG services on Python + FastAPI.
- Developed RAG systems with LlamaIndex: embedding fine-tuning, retrieval orchestration, synthetic QA generation, and multi-modal pipelines over text, charts, and research papers, backed by a FAISS vector store; engineered a long-document RAG path (recursive + sequential summarization) with NeMo Guardrails intent routing, holding time-to-first-token under 5 seconds.
- Built an LLM evaluation framework integrating Ragas (context relevancy/recall, faithfulness, answer relevancy) and LangSmith for tracing and continuous evaluation; added MT-Bench (LLM-as-judge), AlpacaEval, LM-Eval, Elo, perplexity, and a concurrency/latency benchmark measuring per-token timing under load.

--CV

- Built an in-house visual-prompt detector (region-prompt contrastive alignment) that adapts to new domains from a few reference images, with no re-training or re-labeling — +3 AP over YOLOE.
- Fine-tuned OpenVLA for task-specific robotic manipulation and quantized the model for edge inference, cutting memory footprint and latency enough to run the policy on-device rather than off-loading to a server.
- Deployed real-time video analytics on NVIDIA Jetson edge devices across 10+ camera feeds / 5+ sites, sustaining 25–35 FPS with <150 ms latency for tracking, counting, and behavior analytics.

- Built PPE compliance detection (Aramco), achieving 94% precision / 90% recall and reducing false alarms by 25% via post-processing and threshold calibration.
- Built CCTV violence detection (spatio-temporal modeling), reaching F1 ~84% from only 10 minutes of labeled data, cutting annotation time by ~80% via a semi-automated labeling workflow with active-learning data selection.
- Developed thermal/IR image-processing algorithms: fog dehazing/defogging to restore visibility in low-visibility conditions, and optical-flow-based detection of anomalous motion on IR camera feeds.

Technical Lead

IDEfy (Digital Identity/eKYC)

October 2022 — February 2024, Giza, Egypt

- Led digital identity onboarding products (SaaS/SDK/web/mobile), delivering 3 production deployments end-to-end.
- Built validation and quality-control workflows for OCR, face matching, liveness, and identity-verification datasets, improving verification success rate by 20% through data quality tuning and systematic failure analysis.
- Developed configurable OCR/document-processing pipelines that adapted to new structured documents using limited samples, annotation rules, and validation checks, reaching 96% recognition accuracy.
- Analyzed production failure cases across OCR, face verification, and liveness detection to identify noisy data, poor image quality, edge cases, and model retraining opportunities.

Senior ML Engineer (Computer Vision)

RDI

January 2021 — October 2022, Giza, Egypt

- Led the delivery of the Sotoor Arabic OCR System, achieving 12% WER on Arabic manuscripts.
- Built document OCR pipeline modules (layout analysis, segmentation, font detection, recognition, denoising, correction), reducing OCR error by ~7% on noisy scans.
- Adapted the K2 Toolkit for OCR training/decoding, improving training efficiency by ~25% and gaining ~4% WER, boosting sequence recognition stability.

ML Researcher

RDI

January 2017 — January 2021, Giza, Egypt

- Collaborated on a team building NLP systems for the Arabic language across speech and text.
- Developed and deployed production systems, including Speech Recognition, Optical Character Recognition, Text Diacritization, Voice Conversion, and Quran Recitation Verification.

PROJECTS

Agentic Agent Call Center

- Built an agentic AI call-center assistant using LLM tool-calling and Multi-RAG retrieval to automate intent handling, answer generation, and CRM/ticketing actions.
- Designed a reliable conversation orchestration layer with guardrails, escalation-to-human routing, and quality evaluation to ensure safe and consistent customer support automation.

Fashimi APP — Lead Engineer

- Built an AI try-on platform using LLMs for recommendation + prompt generation and diffusion models for avatar generation and photorealistic try-on rendering.
- Developed a recommendation engine leveraging mood/occasion/climate/user preferences to produce personalized outfit prompts.

Valorant Overlay — Developer

- Designed a real-time multi-threaded system extracting gameplay data from Valorant for esports competition overlays.
- Implemented object detection, pattern recognition, and color analysis, achieving a response latency of 50–72 msec.

EDUCATION

Bachelor of Science in Engineering

Cairo University — Electrical Electronics and Communication Engineering · 2016

PUBLICATIONS

**Arabic Multi-Genre Broadcast Speech Recognition** — Conference Paper, December 2019 — IEEE ASRU 2019

**Prognostic AI for Post-Operative Infections** — Conference Paper, November 2023 — FCI Annual Conference

**QuranScript Verification System** — Conference Paper, November 2016 — Sixteenth ESOLE Conference of Language Engineering

HOBBIES

Swimming

Gaming

Padel